



College of Agricultural &
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UNIVERSITY OF GEORGIA

Multi-Locus Typing and Genotyping of *Histomonas meleagridis* Isolates

**Bilic, Jaskulska, Souillard, Liebhart & Hess
PLOS ONE, 2014**

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Histomoniasis

THE PATHOGEN



*Histomonas
meleagridis*

THE TARGET



Devastates turkey
flocks

CONSEQUENCES



High mortality rate

Treatment options are not available; therefore, understanding the genetic diversity is an important step toward tracking and studying this parasite

Why Old Tracking Methods Failed

THE OLD METHOD

ITS region to track
parasites

FLAWS

Intra-isolate heterogeneity
Clonal cultures yielded multiple,
conflicting ITS sequence
variants

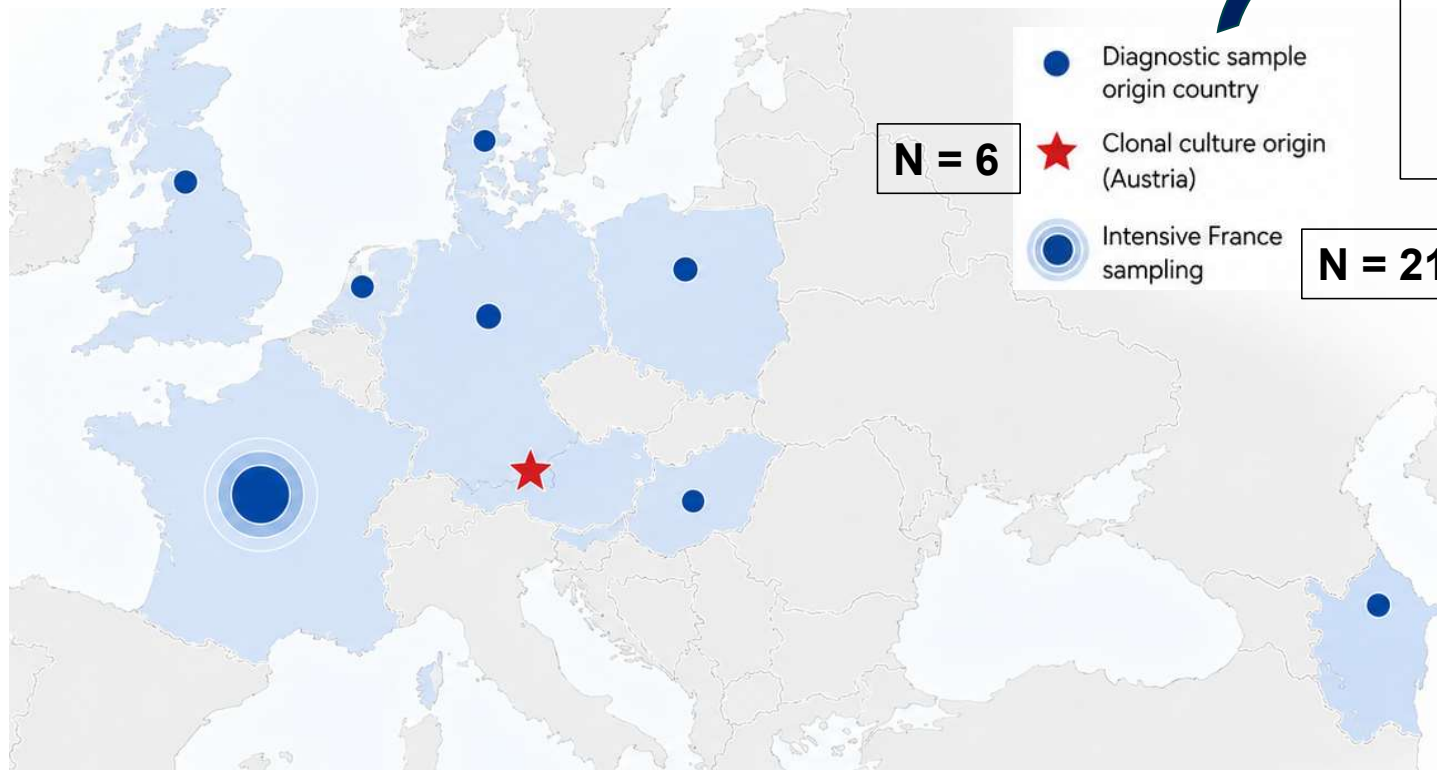
Overlapping Sequences
Identical sequence variants
among completely separate
isolates

*An identity tag is useless if it changes constantly.
ITS markers showed limitations for reliable subtyping.*

THE NEW METHOD

The Multi-Locus Investigation

The Multi-Locus Investigation



Azerbaijan, Denmark,
France, Germany,
Hungary, the Netherlands,
Poland, and the
United Kingdom
N = 39

Total samples: 256

Geographic origin of the samples

Workflow

DNA extraction and
PCR amplification

Sequencing

Sequence Analysis

18S rRNA

Targeting broad genetic
separation

rpb1

Protein-coding, single-
copy candidate

alpha-actinin1

Protein-coding, single-copy
candidate

Assemble + align sequences



Assembly: Accelrys Gene and
Lasergene
Alignment: **ClustalX v2.1**

Confirm identity



BLAST used especially for 18S rRNA
products because the 18S assay
amplified related protozoa as well.

Build phylogenies



PHYLIP v3.68:
Neighbor-Joining
Maximum likelihood

Test tree stability



Bootstrap

1000 replicates for distance trees
100 replicates for ML trees

18S rRNA

Diagnostic Findings

Diversity Score: High. Yielded 34 total sequence types

Identity Range: 95.6-99.8% identity between sequences.

Strengths

Both NJ and ML analyses showed two major *H. meleagridis* groups

Limitations

- Useful for broad separation
- Ultra-fine typing should be interpreted cautiously

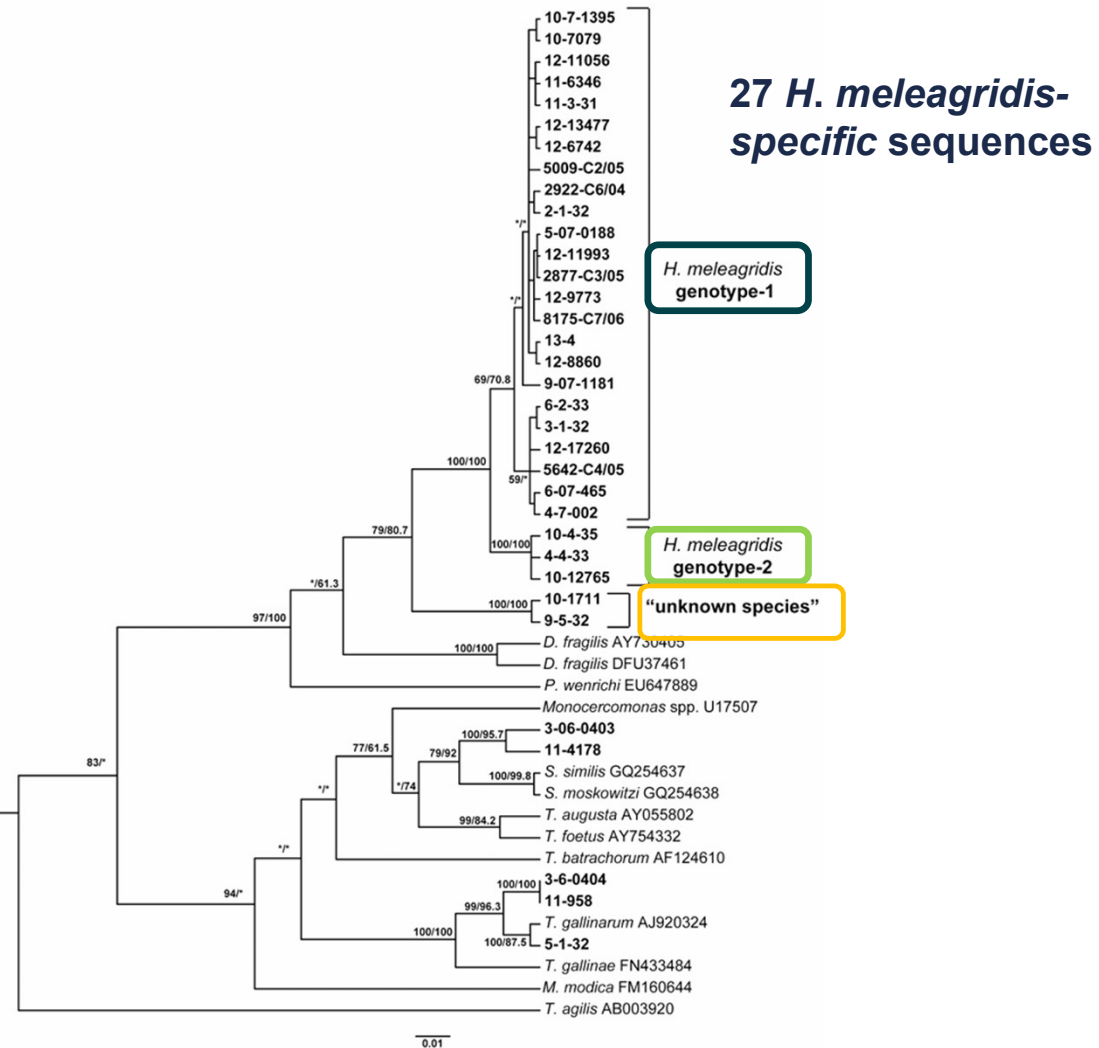


Figure 1. Phylogenetic tree of *H. meleagridis* isolates based on partial 18S rRNA sequences (approx. 544bp).

rbp1

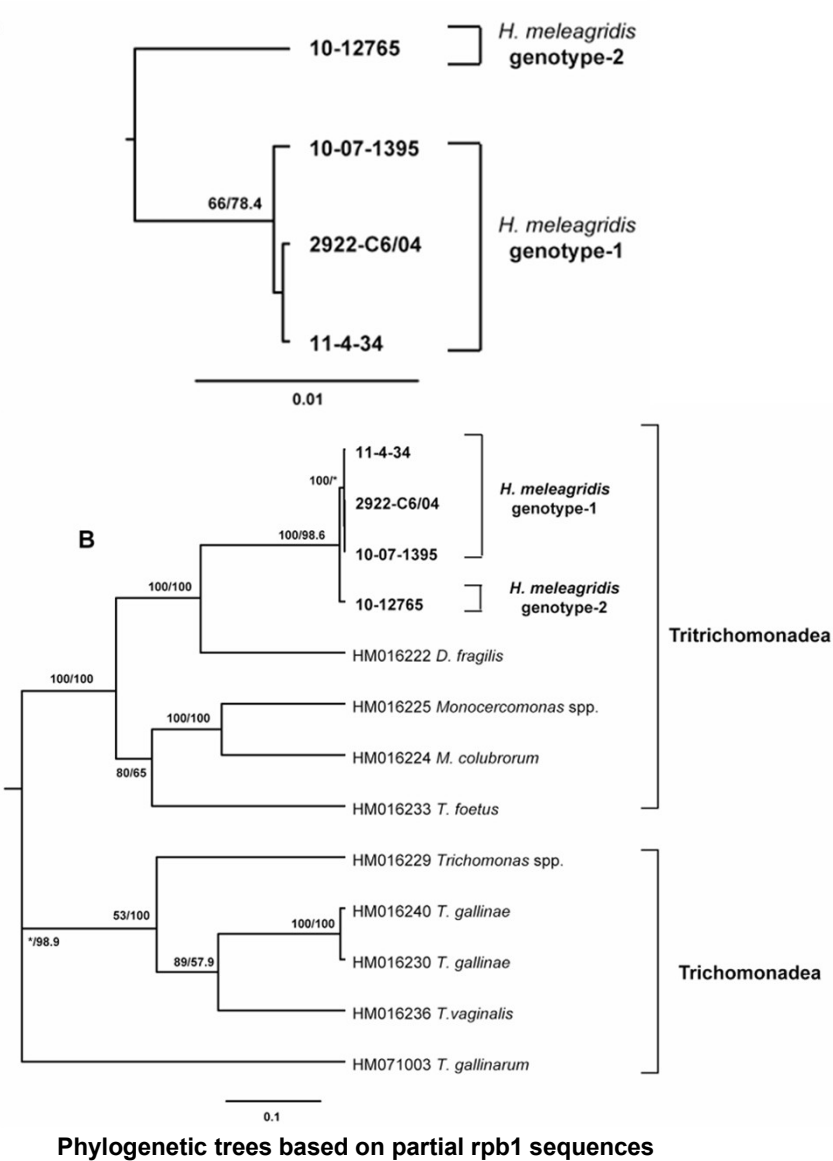
Diagnostic Findings

Diversity Score: Low. Yielded 4 total sequence types.

Differentiation: 17 stable SNPs perfectly separate Genotype 1 and Genotype 2

Adds Resolution
Able to differentiate isolates that the 18S marker identified as identical.

Adds Resolution
Able to differentiate isolates that the 18S marker identified as identical.



alpha-actinin1

Diagnostic Findings

Diversity Score: Low. Yielded 3 total sequence types.

Differentiation: 12 stable SNPs between the two clusters, but very high conservation among samples.

Limitations

99.9% identical within Genotype 1.
100% identical within Genotype 2.

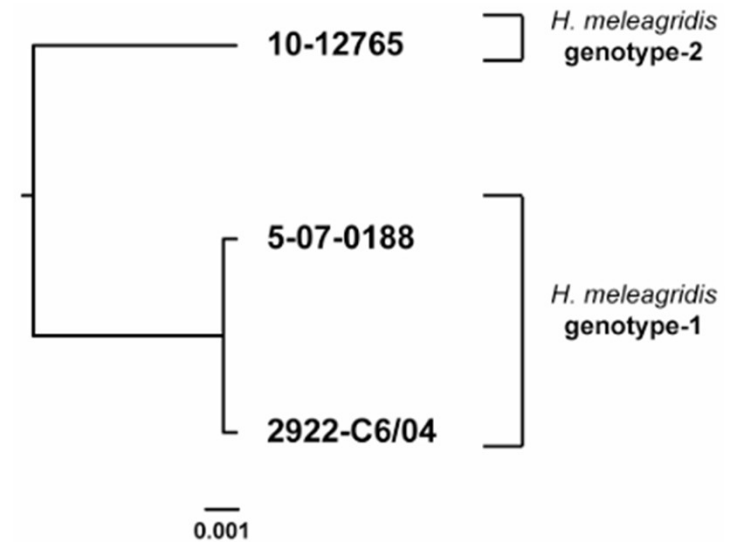


Figure 2. Phylogenetic tree based on partial α -actinin1 sequences of *H. meleagridis* (approx. 1048bp). The phylogenetic

Useful for confirmation, weak for detailed subtyping

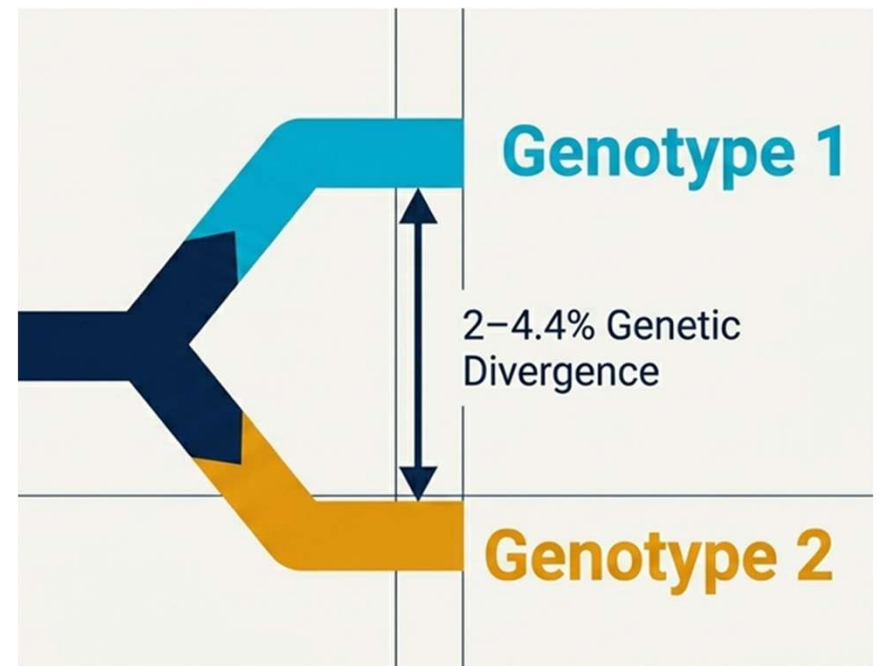
One Parasite, Two Genetic Identities

Phylogenetic analysis across all three tested loci demonstrated a consistent separation of *H. meleagridis* isolates into two distinct clusters.

The Genotype Designation

The genetic variability (2-4.4% difference) mirrors the divergence seen in related protozoa.

Therefore, these are not just clusters; they are two distinct **Genotypes**.



SUMMARY

The Flaw



ITS markers not suitable for tracking *H. meleagridis* due to intra-isolate heterogeneity.

A single cell produces conflicting genetic tags.

The Finding

Genotype 1



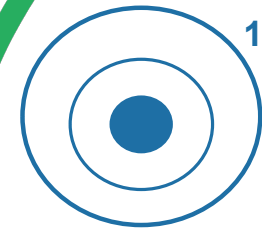
Genotype 2



The parasite exists as two reproducible genetic clusters:

Genotype 1
(widespread across Europe) and **Genotype 2**
(highly localized to France).

The Fix



Combine *18S rRNA* and *rpb1* loci for accurate molecular subtyping

NEXT STEPS

Isolate and Purify

Isolate and purify
Genotype 1 and
Genotype 2

In Vivo Testing

Test the disease
presentation and
mortality associated with
these two genotypes





Any questions

